

Answers

VARC

1. B	2. D	3. B	4. D	5. A	6. A	7. 2	8. C
9. D	10. C	11. A	12. C	13. B	14. B	15. D	16. C
17. C	18. C	19. 1	20. C	21. C	22. C	23. A	24. C

LRDI

25. B	26. 3	27. A	28. D	29. 3	30. 5	31. 12	32. 24
33. 12	34. C	35. A	36. C	37. B	38. A	39. A	40. A
41. 0	42. B	43. 34	44. 42	45. D	46. 10		

Quant

47. C	48. B	49. 70	50. C	51. B	52. D	53. A	54. 2
55. 9	56. A	57. A	58. 2	59. A	60. 30	61. D	62. 8
63. C	64. B	65. B	66. 90	67. 378	68. C		

Explanations

VARC

1. **B**

The given sentence focuses on the impact of these rising costs on employees - an idea that hasn't yet been introduced here. Placing this in Blank 1 would prematurely shift the focus from a general economic observation to the employee-specific consequences, disrupting the logical flow. Similarly, Blank 2 would be a poor choice since it would interrupt the argument about wages not keeping pace with inflation. The broader context of inflation and wages needs to be fully established before narrowing the discussion to employee struggles.

Blank 3 could be a good fit. We observe that the preceding sentence sets up the problem: wages are lagging behind rising costs. The subsequent sentence emphasises the stark contrast between remote work savings and current financial stress; the anecdote from the London University lecturer is further linked to this. Therefore, Blank 4 is also unsuitable. By placing the given sentence in Blank 3, we bridge the two ideas presented above by specifying how these rising costs and stagnant wages directly impact employees. It introduces the tangible pressures employees face, which makes the subsequent sentence about savings during remote work even more impactful.

Hence, Option B is the correct choice.

 VIDEO SOLUTION

2. **D**

Let us evaluate the given choices -

Option A: The creation of new culture and ideas is a central theme of the passage, highlighted as a potential outcome of AI's linguistic capabilities: ["...AI can create completely new ideas, completely new culture..."]

Option B: The author hints that AI could surpass human creativity in areas like storytelling, composing music, and drafting laws or scriptures: ["...What would happen once a non-human intelligence becomes better than the average human at telling stories, composing melodies, drawing images, and writing laws and scriptures?..."]

Option C: The threat to democracy through the mass production of fake news and political content is a major concern raised by the author: ["...Think of the next American presidential race in 2024, and try to imagine the impact of AI tools that can be made to mass-produce political content, fake-news stories..."]

Option D: The passage does discuss AI's ability to create emotional connections with individuals: ["...form intimate relationships with people, and use the power of intimacy to change our opinions and worldviews..."] However, it does not explicitly connect this capability to 'exacerbating the polarization' of political views. The focus is on fostering fake intimacy to influence opinions, not specifically on worsening political polarization.

Hence, Option D is the correct choice.

 VIDEO SOLUTION

3. B

The author emphasises that language is foundational to human culture and civilisation because it:

- articulates and spreads human values and culture (as noted in [Option C](#)).
- lays the groundwork for creating cultural artefacts through storytelling and laws (as noted in [Option D](#)).
- influences political views and fosters emotional ties (as noted in [Option A](#)).

Option B, however, does not align with the rationale behind the "operating system" metaphor: the author does not suggest that language is the "operating system" of civilisation because it underpins AI tools. Instead, the passage treats AI tools like ChatGPT as leveraging language's existing role in civilization.

 VIDEO SOLUTION

4. D

Let us evaluate the choices based on the information in the passage -

Option A: The author does not dismiss fears about students using AI but instead deems such concerns possibly trivial compared to AI's larger societal threats.

Option B: The author doesn't argue that fears of AI harming humans physically are unfounded, but shifts focus to the linguistic and cultural dangers AI presents. It's unclear whether he will explicitly support the view presented here.

Option C: Though the author briefly acknowledges that AI can be used for good, this idea is not a central focus of the passage; he also does not emphasise technological or industrial benefits. Therefore, it's unclear whether the author will support his view.

Option D: The concern stated here has been clearly underlined in the passage. We are informed of the threat AI poses to democracy through its manipulation of language and ability to generate misinformation, fake intimacy, and propaganda. The author explicitly states that democracy relies on meaningful human conversations, which are undermined when AI becomes indistinguishable from humans. Therefore, this option aligns most closely with the author's argument.

Hence, Option D is the correct choice.

 VIDEO SOLUTION

5. A

The passage clearly highlights the potential risks associated with AI's linguistic capabilities, urging action to regulate its use. While the passage does present hypothetical scenarios and employs rhetorical questions, its tone is primarily grounded in a warning, not sensationalism or mere speculation. This points us towards [Option A](#). The author systematically explains the dangers of unregulated AI tools, particularly their capacity to manipulate language and influence human culture. The warnings are thoughtful and aim to provoke awareness and a sense of urgency without excessive dramatisation. The tone is serious and measured, which aligns with a cautionary style.

The remaining tones do not describe the discussion appropriately. For instance, consider [Option B](#): though the passage discusses AI's future implications, the focus is less on prediction and more on warning about what could happen if action is not taken. "Prescient" implies a focus on foresight and vision, but the passage emphasises immediate concerns and actionable advice. Similarly, the passage is not "alarmist" ([Option C](#)), as it avoids overly exaggerated or emotional claims. It uses logical arguments and examples rather than fearmongering. [Option D](#) is also a poor fit: although the passage concludes with a rhetorical question, this is a stylistic device rather than a defining characteristic of the tone. The primary goal is to issue a warning, not to leave the reader in a state of curiosity or wonder, making "quizzical" an incorrect characterisation.

Hence, Option A is the best choice.

 VIDEO SOLUTION



6. A

A useful strategy for determining where the given sentence best fits is to look for places where the flow of ideas feels awkward or disconnected. When we examine the sentences around Blank 1, we see that they are well-linked, moving smoothly from the broader issue of the drought to its immediate consequence, food insecurity. The same logic applies to Blanks 2 and 3 - there are no noticeable disruptions in the flow of ideas here. The structure remains clear and coherent even without inserting the given sentence into these blanks.

However, when we look at Blank 4, we notice a discrepancy. The sentence before Blank 4 mentions farmers and pastoralists, and the sentence following it continues discussing pastoralists, but there's a slight gap in the connection. If the sentences were closely linked, the author could have used pronouns or rephrased the second sentence to refer to pastoralists more naturally. Additionally, there's a shift in focus: the paragraph moves from talking about "leaving them on the brink of famine" to "never being able to go back as their livelihoods have been irreversibly wiped out." This jump in ideas suggests the need for a transitional sentence to bridge the two concepts - enter the given sentence. Moreover, the "Many... Some..." structure works well here. First, the author mentions the larger group of farmers and pastoralists and how they were displaced. Then, the sentence narrows the focus to a smaller portion of this group, highlighting how, even for those who were not displaced, their livelihoods were so affected that they may never recover. This progression makes the insertion of the given sentence at Blank 4 both logical and necessary for maintaining coherence.

Hence, Option A is the correct choice.

7.2

Here, Sentences 1, 4, and 5 discuss the general theme of how non-experts or forecasters without specialized expertise can make accurate predictions, often outperforming experts. Sentence 3 adds context by highlighting that this success occurs even when experts have access to classified information.

Contrarily, Sentence 2 shifts the focus to the “intelligence community’s interest in forecasting models” rather than continuing the discussion on the accuracy and success of forecasters versus experts. This makes it unrelated to the main flow of the paragraph.

Let us examine the points presented in each sentence to further understand how the statements link. Sentence 1 introduces the main idea that forecasting is appealing because it works even without requiring specialized expertise. Building on this idea, Sentence 5 highlights that non-experts can often outperform experts in forecasting; this, in a way, relates to the appeal of forecasting introduced in the first sentence. Sentence 4 provides evidence to support the claim made in Sentence 5. It refers to a specific study where non-experts outperformed experts in predicting geopolitical events, solidifying the argument that non-expert forecasting can be more reliable. Sentence 3 strengthens the previous claim by highlighting that the experts, despite having access to classified intelligence, were still outperformed by non-experts. In this manner, the arrangement 1-5-4-3 renders a coherent paragraph.

Hence, Sentence 2 is the odd one out.

8.C

The statement in the question draws a comparison between Moutai’s claimed hangover-proof quality and a groundbreaking historical invention (gunpowder). The use of “would make it” suggests a hypothetical scenario, and the comparison is meant to emphasise significance, not an actual invention. This indicates that the comparison is metaphorical/figurative (Option C), conveying the liquor’s potential impact on culture and society; hence, we can eliminate Option A.

Option B also implies that the comparison is based on actual substance or tangible qualities. Though the claim about being hangover-proof is substantive, the comparison to gunpowder is not grounded in tangible, measurable terms but rather in its significance. Similarly, Option D suggests that the phrase equates Moutai directly with gunpowder, which is incorrect: the phrase does not treat the two as synonyms but draws a symbolic comparison.

Hence, Option C is the correct choice.

9.D

To identify the factor that is both a reason for Moutai’s success and a potential threat, we could consider aspects that currently drive demand while also holding the potential to hinder it in the future. Looking at the given choices, Option A may be a contributing factor to Moutai’s success, as cultural practices around drinking play a role in its popularity. However, the passage does not explicitly frame this as a threat, especially since it is deeply ingrained in Chinese society, making this option less fitting. Option C is also a significant reason for Moutai’s success, as it targets the super-wealthy, but the passage does not indicate that this would become a threat in the future. Option B is presented more as a hurdle to Moutai’s success, while the explicit benefits remain to be discussed.

On the other hand, we can deduce that Option D is both a key reason for Moutai’s success - by tapping into the spending power of older consumers - and a potential threat, as the younger generations, with different health-conscious lifestyles, could move away from the heavy drinking culture that has driven Moutai’s demand. The passage specifically highlights this generational shift as a risk to Moutai’s long-term success.

Hence, Option D is the best choice.

10.C

The author uses the phrase “the unholy trinity” to describe Moutai’s marketing strategy, which relies on three factors: nationalism, elitism, and ageism. The word “unholy” implies that these factors may be unconventional or controversial, which could be why they are described as such. The author presents this idea in the context of Western marketing, suggesting that Moutai might have succeeded due to these factors, even though they seem to defy conventional Western norms. Thus, the phrase reflects how Moutai’s marketing strategy is in stark contrast to Western business practices (Option C).

Contrarily, Options A, B, and D either do not address the contrast with Western strategies or misinterpret the focus of the passage. For instance, Option A is not the best choice because the phrase “unholy trinity” isn’t a comment on the morality of marketing liquor itself but rather on the controversial nature of Moutai’s specific strategy. Options B and D focus on tangential aspects that are irrelevant to the question.

CAT Percentile Predictor



11.A

Let’s evaluate the given choices and check if they are consistent with the information in the passage -

Option A: According to the passage, this is not true. The author emphasises that Moutai deliberately avoids targeting the middle class, as the competition in that market is intense, and instead focuses on the super-rich. This suggests that marketing to the middle class is not as lucrative or straightforward as it might appear.

Option B: The passage mentions that the Chinese government is Moutai’s largest shareholder and potentially plays a role in controlling the prices of products, which is a key factor for any firm in this market to consider. Furthermore, the author discusses how the government might further interfere with the operations in this space: [“... in what some see as a travesty of corporate governance, its majority owner has plans to set up its own sales channel ...”]

Option C: The author highlights that the premium market in China is still growing and not overcrowded with luxury brands, indicating a potential opportunity for firms targeting high-end consumers.

Option D: The passage states that many firms have failed in their attempt to cater to the middle class due to fierce competition. This is a valid consideration for any firm looking to enter the market.

Hence, Option A is the correct choice.

12. C

To determine where the given sentence fits best, we must consider the logical flow of ideas between sentences. The sentence in question provides important historical and contextual information about the photograph. The author begins by describing the image and ends the paragraph by mentioning the official title of the photo. While sentences offering historical context are typically placed at the beginning, inserting the sentence in either Blank 1 or 2 would create a disruption in the flow of the image's description. This suggests that the sentence must fit in either Blank 3 or 4. Offering contextual information after mentioning the title at the end seems odd; thus, we can eliminate Blank 4.

When we refocus on Blank 3, we see that by this point in the paragraph, the author has just described the soldiers from the South Vietnamese Army's 25th Division following behind the children. This sets up a moment of helplessness that emphasises the chaos and brutality of the scene. The sentence about the trauma and violence of the war logically follows the description of the soldiers because it explains the broader implications of the photo's content, providing a context for the violence and its far-reaching consequences. After discussing the soldiers, the sentence connects the personal and immediate suffering in the photograph with the larger scope of the Vietnam War's impact, which then leads into the official title of the photograph. The official title, "The Terror of War," is directly linked to the broader theme of violence and destruction that the sentence describes. Thus, the given sentence naturally sets the stage for introducing the official title and the iconic nickname, "Napalm Girl," ensuring a smooth and coherent transition.

Hence, Option C is the correct choice.

13. B

The question asks us to identify the hypothetical scenario that does not reflect the kind of loss described in the passage. The passage discusses cultural, ecological, and intellectual losses caused by language extinction, including the loss of:

- Unique cultural expressions (e.g., music, art, and emotions tied to language).
- Knowledge about the environment (e.g., relationships between plants and illness).
- Worldviews and philosophical insights.

We need to identify a scenario that doesn't align with these themes. Let us examine the options based on this understanding -

Option A: This scenario reflects the loss of unique cultural knowledge - in this case, the ability to describe the "20 different moods of the ocean," which likely represents detailed ecological and environmental understanding.

Option B: This scenario focuses on some form of administrative or statistical change in a government list; the focus is not on the cultural, ecological, or intellectual loss emphasised in the passage. Therefore, the option does not reflect the deeper, humanity-wide loss described in the passage.

Option C: This scenario aligns with the loss of unique cultural concepts and the emotional depth tied to a word or phrase. In the author's perspective, losing this concept would diminish humanity's understanding of the complexity of human relationships.

Option D: This scenario reflects the loss of ecological knowledge about snow textures, which likely has practical implications for living in the Arctic environment. This aligns with the passage's discussion of losing environmental wisdom when languages die.

Hence, Option B is the correct choice.

14. B

The central idea of the passage is that endangered languages, as carriers of unique cultural perspectives and human knowledge, must be preserved to benefit humanity. The author argues that liberal arts education plays a vital role in this preservation, both by fostering global citizenship and by encouraging some individuals to directly engage in preservation efforts. Liberal arts education, as framed in the passage, sensitises students to cultural diversity and equips them with the tools to support endangered languages and cultures.

We observe that Option B, however, directly undermines this central idea by redefining the focus of liberal arts education. Requiring fluency in two of the most widely spoken global languages (e.g., English, Spanish, Mandarin) would prioritise dominant languages rather than endangered ones. This hypothetical scenario shifts resources and attention away from the preservation of linguistic diversity, which is central to the passage's argument. Such a requirement would reinforce the dominance of already powerful languages, the very phenomenon identified as a major "language killer" in the passage. By institutionalizing the focus on dominant languages, it would erode the argument that liberal arts education fosters support for endangered languages and cultural preservation, ultimately weakening the role of liberal arts in addressing language extinction.

In contrast, the remaining choices either align with the discussion or do not serve as strong counterarguments to the points presented in the passage. For instance, Option A acknowledges that most liberal arts students will not directly engage in language preservation but does not challenge the broader idea that liberal arts education fosters empathy and support for endangered languages. Option C limits the long-term success of language preservation but does not negate its immediate benefits or the potential for renewal in subsequent generations. Similarly, Option D highlights a limitation of recording dying languages but does not undermine the broader argument that documentation is a valuable and necessary tool in preservation.

15. D

The passage highlights the widespread extinction of indigenous languages in North America due to colonisation, physical annihilation, and assimilation policies while noting slightly better survival rates in Central and South America. It implies that language survival may be influenced by factors such as social policies, cultural integration, and the extent of physical and cultural displacement.

Evaluating the choices, we note that Option A is plausible since allowing children to stay with families would help preserve native languages, unlike the North American policy of removing children to boarding schools, as discussed in the passage. Option B is also reasonable, as less effective assimilation efforts by colonial governments could lead to better language retention. Option C also aligns with the passage's context, as it discusses physical annihilation as a significant driver of language extinction, but the survival rate being slightly better in South America could suggest marginally less physical annihilation.

However, Option D is problematic; while providing locals with jobs in the colonial administration might seem like a factor that supports language retention, this scenario is not consistent with the passage. The passage emphasises that dominant languages often replace indigenous ones through socio-economic pressures, and employment in colonial administration would likely reinforce the use of the dominant language rather than preserve native languages.

Hence, Option D is the best choice.



16.C

The author discusses how a liberal arts education, combined with participation in language preservation efforts, empowers students in several significant ways. He highlights that such an education broadens students' cultural understanding, helps them communicate across linguistic barriers (Option A), and enables them to gain insights into both their own and others' cultures (Option D). Option B is an implicit aspect of this discussion. The passage also suggests that some students may even become involved in active language preservation, though it emphasises that the majority might not pursue this path. Contrarily, Option C is not explicitly mentioned or understood: while the passage provides examples of communities that have established such schools, it does not suggest that students themselves will take on this role.

[VIDEO SOLUTION](#)

17.C

Option C is the correct answer. This option best captures the essence of the passage. The tradwife's embrace of traditional roles and vintage fashion naturally contrasts with and highlights the superficiality of modern life. The passage emphasizes that she is not overtly critiquing society but rather embodying a regressive ideal that challenges contemporary norms. This fits with how the passage describes her actions—her existence itself is a challenge to modern societal expectations.

Option A: The passage does not suggest that the tradwife is actively critiquing modern society. Instead, the tradwife embodies a lifestyle and set of ideals that are in contrast to contemporary values. The critique comes from others' reactions to her behavior, not from her intentions.

Option B: The passage does not show her as actively trying to expose or mock anything. She simply lives in a way that contrasts with modern expectations. The "mockery" is a reaction from others, not her goal. Therefore, this option overstates her intentions.

Option D: This focuses on the tradwife's dress and adherence to traditional roles as the means of revealing modern life's artificial nature. While the tradwife does embody these traditional values, the passage is more focused on how her actions highlight societal superficiality rather than "revealing" it. Additionally, the passage suggests that her behaviour is not an overt revelation but rather something that others react to—making this option somewhat distorted.

[VIDEO SOLUTION](#)

18.C

The passage contrasts the personal, introspective nature of lyric poetry with Marxism's outward focus on societal change. At first glance, the characteristics of lyric poetry - being personal, small-scale, and idiosyncratic - seem incompatible with Marxism's demand for a transformative critique of an unjust society. However, the passage introduces a nuanced perspective within Marxist thought, which recognises lyric poetry as having an implicit critical and utopian function. Through its imaginative and alternative modes of perception, lyric poetry can challenge dominant ideologies and suggest a vision of resistance to oppression. Thus, while lyric poetry does not directly engage with large-scale social concerns, it can still align with Marxist ideals by offering subtle forms of critique and hope for change. Option C most effectively captures this idea.

None of the other choices present a valid interpretation of the passage. For instance, Option A oversimplifies the issue by portraying lyric poetry as merely 'utopian,' ignoring its critical and resistant potential as described in the passage. It misses the nuanced compatibility suggested between lyric poetry and Marxism. Option B inaccurately suggests that Marxism has "internal contradictions," which is not the focus of the passage. Similarly, Option D frames Marxism as dismissive of lyric poetry, ignoring the strand of Marxist thought that finds value in poetry's alternative perspectives; this misrepresentation makes it a weaker summary.

[VIDEO SOLUTION](#)

19.1

A good starting point here would be to link sentences 5 and 3 based on the idea of "pruning." Sentence 5 talks about the synaptic development that occurs during the early period described in sentences 4 and 2. It describes how synapses are initially overproduced, a concept that sets up the next part of the process: synaptic pruning. Sentence 3 continues this discussion by offering an explanation for why synapses are pruned: to reinforce the necessary connections and eliminate the weaker ones. This helps to fine-tune the brain's wiring, ensuring that the important pathways are retained while the less useful ones are discarded. Together, 5-3 ties in with the broader purpose of neural wiring presented in Sentence 2: it emphasises that the creation of neural connections in the fetus is a preparatory stage and that these connections are essential for the future performance of the visual system. Sentence 4 sets the context of the paragraph: it mentions the formation of neural connections before birth, specifically focusing on the connections between the eyes and the brain.

While Sentence 1 provides a technical description of synapse creation and function, it doesn't fit the broader context set by sentences 4, 2, 5, and 3. These sentences focus on brain development (especially related to vision) and the process of synapse formation and pruning in a developmental context. Sentence 1 focuses on the structural aspect of synapse formation (the "specialized structures" at the terminal end of an axon), which is more about how a synapse functions at a microscopic level rather than the larger developmental or functional role synapses play in early brain development, as discussed in the other sentences.

Therefore, Sentence 1 is the odd one out.

[VIDEO SOLUTION](#)

20.C

The passage discusses the challenges of regulating innovations in biotechnology. It highlights two key issues: (i) current regulations are outdated and insufficient for new biological advancements, and (ii) the unpredictable nature of living entities and rapid technological changes make it difficult to design a regulatory framework that anticipates all risks and contingencies. The core question is whether regulation can ever keep pace with the unpredictable and unique outcomes of biotechnology. Option C correctly touches upon these ideas.

The other options are either too narrow or miss the essence. Option A focuses on calibration and scientists' roles, which is not the central concern of the passage. Similarly, Option B highlights the urgency for new rules but omits the skepticism about achieving comprehensive regulation. Option D overemphasises the impossibility of imagining all risks without addressing the need for flexibility in regulations.

Hence, Option C is the best choice.

[VIDEO SOLUTION](#)

21. C

The phrase “unlikely to disagree + EXCEPT” can seem tricky to interpret. In simple terms, the question requires us to find a statement the author will disagree with. Let us inspect the choices -

Option A: The author supports the proposal for zonal segregation as a reasonable compromise, balancing scientific exploration with human settlement.

Option B: The author agrees that NASA’s earlier missions did not prioritise contamination but implies they caused no significant harm.

Option C: This viewpoint reflects a cautious approach to space exploration. The author dismisses concerns about hypothetical extraterrestrial life as speculative and prioritises human exploration and development over minimising contamination. Therefore, he’s likely to disagree with this position.

Option D: In the passage, the author argues that the costs of maintaining strict planetary protection measures are excessive and could undermine future exploration efforts. This is consistent with his stance.

Hence, Option C is the best choice.

VIDEO SOLUTION

22. C

The passage discusses the debate surrounding planetary protection policies, particularly the concerns about contaminating Mars with Earth-based microbes. The author argues against these concerns, citing several reasons why the risk of contamination should not hinder human exploration and development of Mars. These reasons include:

- the lack of evidence for life on Mars (describes Mars as a “bleak, rusted landscape” with no confirmed life) [Option A]
- the disregard for such protocols by international competitors (China’s lenient approach to planetary protection) [Option B]
- the historical precedent of contamination from earlier human missions (Apollo missions left waste on the Moon) [Option D]

On the other hand, Option C is not presented as a valid reason. The author does not specifically argue that probes have had “little effect” on the Moon’s environment but instead focuses on human waste and contamination from earlier human missions, not robotic probes.

VIDEO SOLUTION

23. A

The first paragraph critiques the stringent planetary protection policies advocated by a group of scientists who aim to prevent biological contamination of celestial bodies. The author portrays these efforts as excessive, particularly given the lack of evidence for extraterrestrial life, and highlights the significant financial burden these measures place on space agencies like NASA. Option A accurately reflects this scepticism, as the author questions the need to sterilize planets where life has not been proven to exist.

Option B is incorrect because the author is not equivocal (i.e., ambiguous or undecided); instead, he expresses a clear stance against these strict protocols.

Option C is also inaccurate, as the author is not indifferent to elitism but rather critiques the scientists’ restrictive approach. Similarly, Option D can be eliminated because the author does not approve of NASA’s spending on sterilization but views it as an unnecessary expense.

VIDEO SOLUTION

24. C

The passage highlights contrasting reactions to two instances of potential contamination of the lunar environment: China’s germination of a plant seed on the Moon, which elicited little controversy, and Israel’s accidental release of tardigrades aboard the Beresheet probe, which sparked significant backlash within the space community. This contrast underscores differences in how national or regional scientific communities respond to issues of planetary protection. Option C most closely reflects this idea.

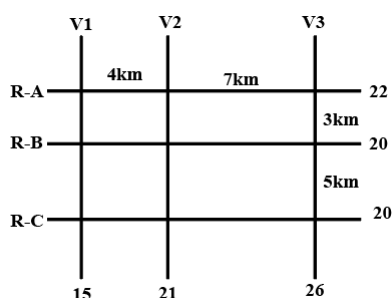
The passage does not suggest that contamination from animals is inherently more harmful than from plants, as suggested in Option A. Similarly, Option B inaccurately implies that the passage endorses China’s approach as inherently “reasonable,” which it does not. Option D diverges a bit from the discussion by emphasising global biases against specific countries, but the passage provides no evidence of such biases, focusing instead on scientific reactions.

VIDEO SOLUTION

LRDI

25. B

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

	V1	V2	V3	
R-A		4km	7km	15
R-B			3km	22
R-C			5km	20
	15	21	26	

We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.

	V1	V2	V3	
R-A	7	4km	7km	15
R-B	8	12	3km	22
R-C		9	5km	11
	15	21	26	

Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATM's on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.

	V1	V2	V3	
R-A	15	4km	7km	7
R-B		12	8	3km
R-C		9	5km	11
	15	21	26	

Using the two cases, we can answer the given questions.

The only statement that is correct considering both the cases is Option B

"The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 9 Lakhs."

[VIDEO SOLUTION](#)



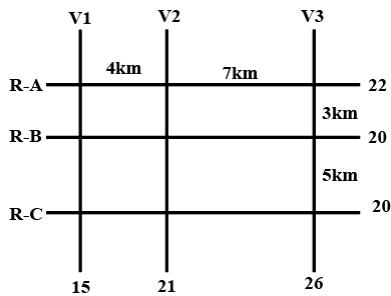
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26.3

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

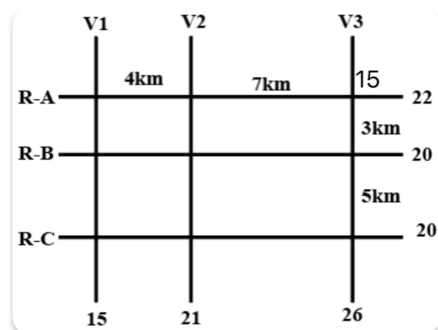
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

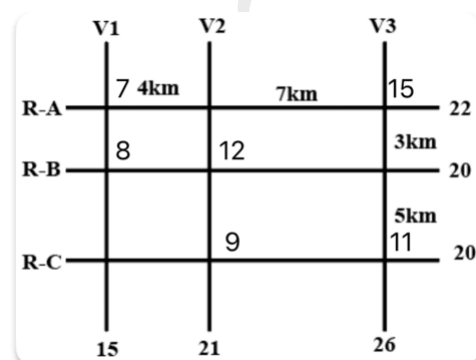
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



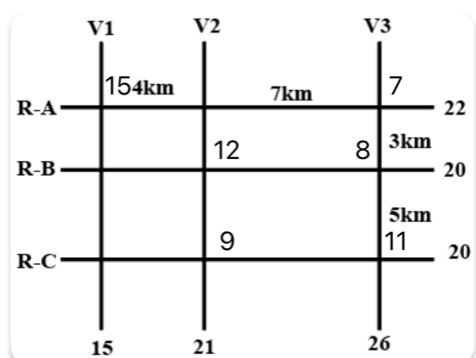
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



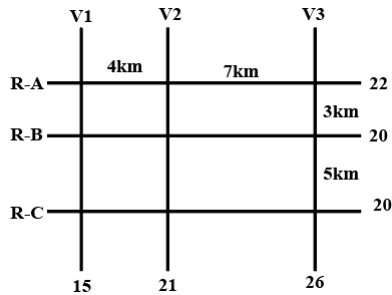
Using the two cases, we can answer the given questions.

Three ATM's with 10L or more: 15L, 12L and 11L. Answer is 3.

[VIDEO SOLUTION](#)

27. A

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

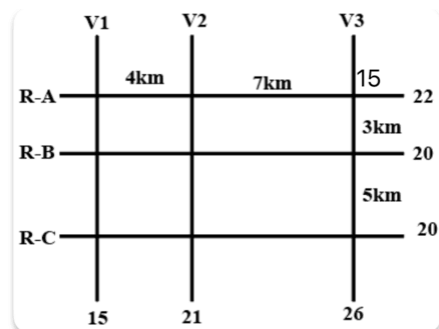
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

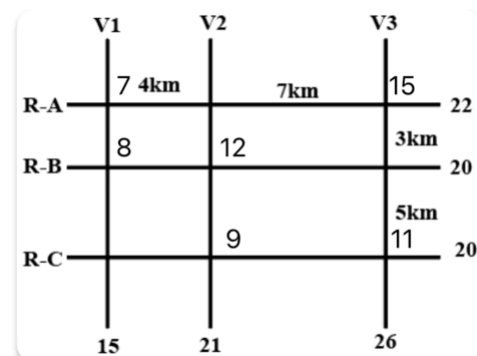
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



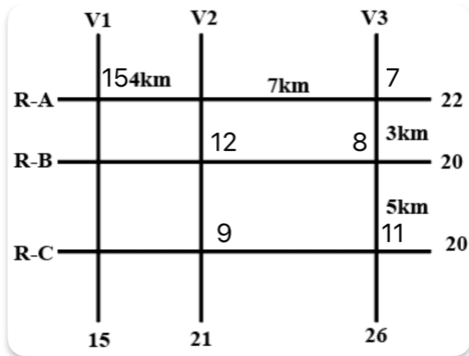
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

We see that in both cases, RA RB and RC have two ATMs

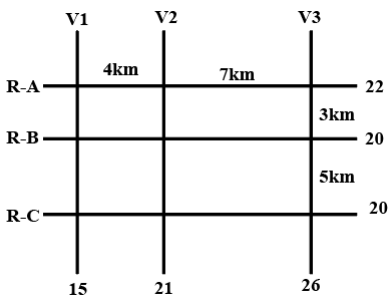
Only in the first case V1 V2 and V3 have two ATM's, in the second case V3 has 3 ATMs and V1 has 1.

Hence only Statement A is correct.

VIDEO SOLUTION

28.D

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

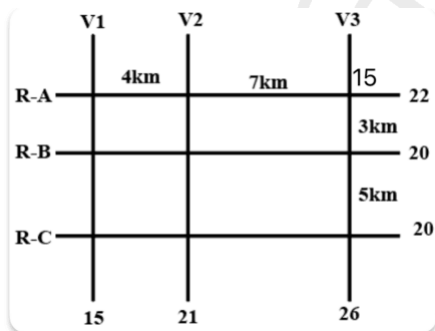
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

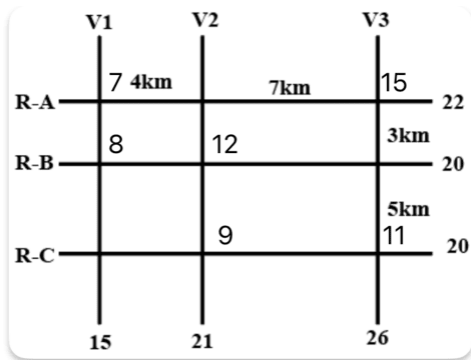
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



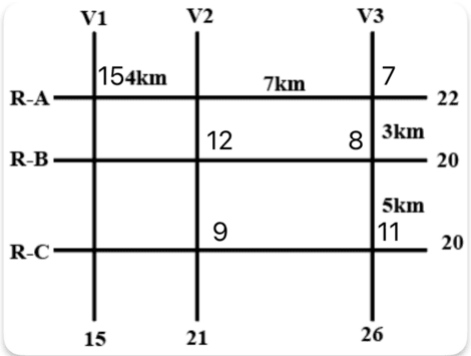
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

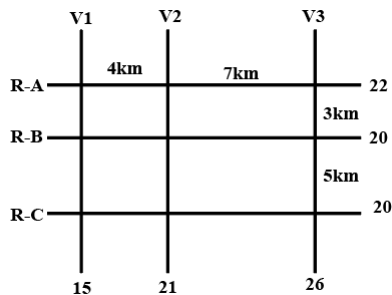
Second highest is 12L and second lowest is 8L

In Case 1 the distance between 12L and 8L ATM is 4km and in Case 2 the distance between 12L and 8L ATM is 7km. Hence the answer is either 4km or 7km.

[VIDEO SOLUTION](#)

29.3

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

	V1	V2	V3	
R-A	4km	7km	15	22
R-B			3km	20
R-C			5km	20
	15	21	26	

We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.

	V1	V2	V3	
R-A	7 4km	7km	15	22
R-B	8	12	3km	20
R-C		9	11 5km	20
	15	21	26	

Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.

	V1	V2	V3	
R-A	15 4km	7km	7	22
R-B		12	8 3km	20
R-C		9	11 5km	20
	15	21	26	

Using the two cases, we can answer the given questions.

ATMs that can be uniquely determined are the ATMs with cash 9L, 11L and 12L. Hence the answer is 3.

[VIDEO SOLUTION](#)

Explanation [30 - 33]:

The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.

We have the reference point of P1 at 66

Trying to fill in for C1:

The possible values are 75, 80, 85, 90, 95

Since 12 grams is of other nutrients, we can eliminate 90 and 95.

If we take 85, $85+12 = 97$, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, $75+12 = 87$ would leave 13 grams of protein, which, too, is eliminated.

Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.

Taking 85 carbs would give 2 grams of protein, which can be eliminated.

80 and 70 would also not work for the same reason.

Leaving only 75 grams of carbs, leaving 12 grams of protein.

30.5

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that all other foodgrains except M3 had more carbohydrates than M1.

Therefore, 5 is the correct answer.

VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile



		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that there were 12 grams of protein in M2.

[VIDEO SOLUTION](#)

32.24

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

There were 24 grams of other nutrients in M3.

[VIDEO SOLUTION](#)

33.12

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

The proteins in the foodgrains, when arranged in ascending order, are: 8, 8, 10, 12, 12, 14, 16

Giving the median as 12.

[VIDEO SOLUTION](#)

Explanation [34 - 37]:

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2024 as 110x
This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

34. C

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

The first question asks about the number of others in 2024, which is 55x, or simply 55,000

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

35. A

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

There are 15x, that is, 15,000 kids in 2023, of which we are given that 10,000 use one app.

So, 5,000 use multiple apps.

The percentage of kids using multiple apps in 2023 would hence be $\frac{5000}{15000} \times 100 = \frac{100}{3} = 33.33\%$

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)



36. C

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

In 2023, there were 20,000 elders; in 2024, there were 33,000 elders.

The increase in percentage would be $\frac{33000 - 20000}{20000} \times 100 = \frac{13}{20} \times 100 = 65\%$

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

37. B

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

In clue 2, we were given that 33000 kids and elders out of 55000 kids and elders use one app, showing that 22000 use multiple apps. To minimize the total usage of multiple apps, we can assume that all 55000 other subscribers use one app.

Giving the percentage of multiple app users as: $\frac{22,000}{110,000} \times 100 = \frac{100}{5} = 20\%$

Therefore, Option B is the correct answer.

Explanation [38 - 41]:

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

38. A

In this question we are asked to find which of the countries among the four options has the least population in 2024 (The current year)

Option A: Country 8

GDP=0.07 and GDP Per Capita=0.41

Population=0.07/0.41 or $\frac{7}{41}$

Option B: Country 3

GDP=0.13 and GDP Per Capita=0.02

Population=0.13/0.02 or $\frac{13}{2}$

Option C: Country 7

GDP=0.08 and GDP Per Capita=0.3

Population=0.08/0.3 or $\frac{8}{30}$

Option D: Country 5

GDP=0.1 and GDP Per Capita=0.36

Population=0.1/0.36 or $\frac{10}{36}$

Comparing the four values, we are looking for the smallest values among the four,

$\frac{7}{41}, \frac{13}{2}, \frac{8}{30}, \frac{10}{36}$

$\frac{7}{41}$ is the smallest value among the four, hence Option A or Country 8 is the answer.

39. A

We are given the GDP growth rate and we are told it is constant over the next three years,

Country 4 GDP: 0.12 and the given growth rate is 0.5%

GDP of Country 4 in 2026 will be $0.12 \times (1.005)^2$ this equals 0.121203

Country 4 GDP: 0.1 and the given growth rate is 3.2

GDP of Country 5 in 2026 will be $0.1 \times (1.007)^2$ this equals 0.1014049

Ratio will be, $\frac{0.121203}{0.1065024}$ this equals 1.19523

Hence, the answer is Option A

40. A

The given question asks us to compare the population of countries between 1, 4, 5, and 7 in the year 2027

Country 1: Population in 2024 will be 0.15/0.41 or 15/41 and the population is decreasing at the rate 0.12%

Country 4: Population in 2024 will be 0.12/0.38 or 12/38 and the population is increasing at the rate of 0.49%

Country 5: Population in 2024 will be 0.1/0.36 or 10/36 and the population is increasing at the rate 0.31%

Country 7: Population in 2024 will be 0.08/0.3 or 8/30 and the population is decreasing at the rate 0.11%

Simplifying and comparing the four fractions in order, $\frac{15}{41}, \frac{6}{19}, \frac{5}{18}, \frac{4}{15}$

The decimal values of the following are: 0.36586, 0.315789, 0.2777, 0.2666

Right away we can eliminate Country 7 since the population is not only the least, but it is decreasing, so there is no chance it will have the highest population in three years time.

For the other three countries,

Country 1: $(0.36585) \times (0.9988)^3$ this equals 0.36453

Country 4: $(0.315789) \times (1.0049)^3$ this equals 0.32045

Country 7: $(0.2777) \times (1.0031)^3$ this equals 0.2803612

Hence, Country 1 will have the largest population among these four countries.

[VIDEO SOLUTION](#)



41.0

We are asked to find the number of countries where the GDP per capita is lower in 2027 than it was in 2024.

For the GDP per capita to be lower in the consequent years, the population growth rate has to exceed the GDP growth rate, since GDP per capita is nothing but GDP divided by the population. That means if the population growth rate is lesser than that of the GDP growth rate, the GDP per capita will only increase.

We can clearly see that none of the following countries fall in the scenario where GDP growth rate is lesser than that of the population growth rate. Countries 1, 2 and 7 have actually decreasing population rates thereby definitely increasing the GDP per capita.

The rest of the countries have GDP growth rates larger than the population growth rates.

Hence, we can conclude that, none of the countries will have a smaller GDP per capita in 2027 when compared to 2024.

[VIDEO SOLUTION](#)

Explanation [42 - 46]:

The set's starting point is the conclusion that the temperature rises only when the AC is turned off.

When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celsius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

42. B

From this alone, we can answer the first question:

The AC was turned off for only two instances between 11:01 pm and 1:59 am:

Once at 0:00 and once at 1:00

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

43. 34

From the table, we can determine that the temperature outside at 1 am was 34 degrees Celsius.

Therefore, 34 is the correct answer.

[VIDEO SOLUTION](#)

44. 42

Extrapolating this linear chain of temperature drop (2 degrees drop every hour), we can see that the temperature at 22:00 would be 40 degrees Celsius, and the temperature at 21:00 or 9 pm would be 42 degrees Celsius.

Therefore, 42 is the correct answer.

[VIDEO SOLUTION](#)

45. D

We can see that the AC was turned on at 0:30, 1:30 and 23:00.

We do not know about the status at 23:30

We are to count the times after 11:01 pm; hence, turning on the AC at 23:00 does not count.

Hence, the answer should be 2 or 3.

But in the final clue, we are given that POWER mode was used longer than REGULAR mode.

Since there are 4 periods of 30 minutes each, the combinations possible are POWER-3, REGULAR-1, and POWER-4, REGULAR-0

If the AC were on REGULAR mode at 23:00, it would have functioned for 1 hour, and the power mode would have also functioned for 1 hour (total), which would go against the final clue.

The following would be the cases.

	Case-1	Case - 2	Case - 3	Case - 4	Case - 5
23:00	Regular	Power	Power	Power	Power
23:30	Power	Regular	Power	Power	Power
0:30	Power	Power	Regular	Power	Power
1:30	Power	Power	Power	Regular	Power

But there were three instances of activation between 11:01 and 1:59

Hence, Option D would be the correct answer.

[VIDEO SOLUTION](#)



46.10

(Side note: We cannot consider the temperatures outside and inside at 23:00 and 2:00 since the time limit does not include those.)

The maximum difference between the elements of the outside and inside is maximum at 0:00 when the outside temperature is 36, and the inside temperature is 26.

Giving the maximum difference as 10

Therefore, 10 is the correct answer.

[VIDEO SOLUTION](#)

Quant

47.C

Total number of students is 250, and we are told that, The percentage of girls was at least 44% and at most 60%.

So the number of girls range from, $0.44(250) \leq \text{Girls} \leq 0.6(250)$

$110 \leq \text{Girls} \leq 150$

Statement 1:

If 50% of the boys and 80% of the girls opted for swimming, that means if the total number of Boys is B, Girls is G where $B+G=250$.

Swimming is: $0.5B+0.8G$

Statement 2:

If 70% of the boys and 60% of the girls opted for running, that means

Running is $0.7B+0.6G$

Total number of enrolments for swimming and running together will be

$(0.7B+0.6G)+(0.5B+0.8G)=1.2B+1.4G$

Using the overlapping principle, where I represents people who have enrolled only for one activity and II represents number of people who have enrolled for two activities.

We know that, $I + II = 250 = B + G$

$I + 2II = 1.2B + 1.4G$

Subtracting the two equations,

$II = 0.2B + 0.4G$

$II = 0.2(B + 2G)$

Using $B+G=250$

$II = 0.2(250 + G)$

G can at-most be 150 and at least 110.

So maximum value of II will be $0.2(250 + 150) = 80$

Minimum value of II will be $0.2(250 + 110) = 72$

[VIDEO SOLUTION](#)

48.B

Opening the square on the left-hand side, we get $a^2 + 3b^2 + 2ab\sqrt{3}$

Comparing the rational part on both side,s we get: $a^2 + 3b^2 = 52$

And comparing the irrational part, we get: $2ab\sqrt{3} = 30\sqrt{3}$

$ab = 15$, Since we are given that a and b are natural numbers, the possible values of a and b are (1,15), (3, 5), (5, 3), or (15,1)

Putting these values in the first relation we got, we see that 15 squared would exceed the required value and would not be the case.

We need not check if $a=5, b=3$ or $a=3, b=5$ since the answer would be the same.

($a=5$ and $b=3$ would satisfy it)

$$a+b = 5+3 = 8$$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

49.70

We are given that average of three distinct integers is 28, that means the sum of these three integers is $28 \times 3 = 84$

Let us write $x + y + z = 84$

x, y, z being the three distinct integers in ascending order.

If the smallest number is increased by 7 and the largest number is reduced by 10

$$(x+7) + (y) + (z-10) = 81$$

New arithmetic mean will be $\frac{81}{3} = 27$

And this is said to be 2 more than the middle number, meaning

$$27 - 2 = y = 25$$

$$x + z = 59$$

We are given that difference between the largest and the smallest numbers becomes 64,

$$(z-10) - (x+7) = 64$$

$$z - x = 81$$

Adding the two equations we get, $2z = 140$

$$z = 70$$

[VIDEO SOLUTION](#)

50.C

There are multiple ways of solving such questions involving remainders; one easy way is to look for a power of numerator that leaves a remainder of 1 or 01 when divided by the denominator.

In this instance, 1000, when divided by 13, leaves a remainder of -1

We can rewrite the numerator as $\frac{10^{66} \times 100}{13}$

The remainder would be $\left[\frac{10^{66}}{13} \right]_R \times \left[\frac{100}{13} \right]_R$

$$(-1)^{22} \times 9$$

9

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.B

We are given that Sam completes a piece of work in 20 days. We are also given that Mohit is twice as fast, so he should take only 10 days. Mohit is thrice as fast as Ayna, so he would take 30 days.

Let's take the total work to be 60 units; this would give the work done per day for Mohit, Sam, and Ayna to be 6, 3 and 2, respectively.

On the first day Sam and Mohit work: doing 9 units

On the second day Sam and Ayan work: doing 5 units

On the third day, Mohit and Ayan work: doing 8 units

Essentially doing 22 units in a 3 days cycle.

After two such cycles, there will be $60 - 44 = 16$ units of work left

On day 7, Sam and Mohit would work 9 units, leaving 7 units

On day 8, Sam and Ayan would work 5 units, leaving 2 units

And on day 9, Ayan and Mohit would complete the remaining work.

So Sam worked for a total of $2+2+2=6$ days and on each day he did 3 units of work, completing 18 units of work.

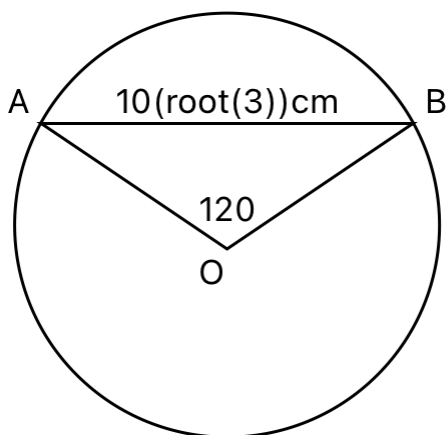
The ratio of work done by Sam would be $\frac{18}{60} = \frac{3}{10}$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

52.D

This is the situation that is described in the question above, Angle AOB is 120 degrees and the chord AB is of length 10cm.



Using Cosine rule we can find the length of AO and AB

$$\cos(\angle AOB) = \frac{r^2 + r^2 - 300}{2r^2}$$

$$-\frac{1}{2} = \frac{(2r^2 - 300)}{2r^2}$$

$$3r^2 = 300$$

$$r = 10$$

Area of the Sector AOB is $\frac{120}{360} \times \pi \times r^2$
 $\frac{1}{3} \times \pi \times \frac{100}{1}$ which is $\frac{100\pi}{3}$

Area of triangle AOB is $\frac{1}{2} \times r^2 \times \sin(120)$

$$\frac{1}{2} \times \frac{100}{1} \times \frac{\sqrt{3}}{2}$$

$$\text{Area of triangle AOB is } \frac{75}{\sqrt{3}}$$

The smaller region will be, $\frac{100\pi}{3} - \frac{75}{\sqrt{3}}$

Taking 25 common we will get,

$$25 \left(\frac{4\pi}{3} - \sqrt{3} \right)$$

[VIDEO SOLUTION](#)

53. A

Finding the terms in the sequence, we see that $t_3 = 0, t_4 = -\frac{1}{3}, t_5 = 0$

We would notice that all the odd terms are 0, and we are also asked the sum of only even terms, so we do not need to consider those

$$t_6 = -\frac{1}{5}$$

We see that the even terms are in an HP: $-1, -\frac{1}{3}, -\frac{1}{5}, -\frac{1}{7}, \dots$

The sum we are asked is the inverse of these terms, that is: -1, -3, -5, -7, up to 1012 terms

The sum of this AP would be $\frac{[-(2 \times 1) + (1012 - 1)(-2)]}{2} \times 1012$

Which is equal to $-1012 \times 1012 = -1024144$

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)

54. 2

The expression on the right-hand side will have two critical points: 2 and -2

For any value of **x greater than equal to 2**, the equation changes to $x+2-(x-2) = 4$

the value of $\min\{x, 2\}$ would be 2, so we would want $\max\{x, 2\}$ to be $4+2 = 6$

Therefore, $x=6$ works.

For any value of **x less than equal to -2**, the equation changes to $-(x+2)+(x-2) = -4$

the value of $\max\{x, 2\}$ would be 2, so we would want $\min\{x, 2\}$ to be 6 again; this cannot be the case.

In general, subtracting the smaller (min) of the two values from the bigger (max) can not lead to a negative number. The max it can lead to is a 0

When **x lies between -2 and 2**, the equation becomes $2x$

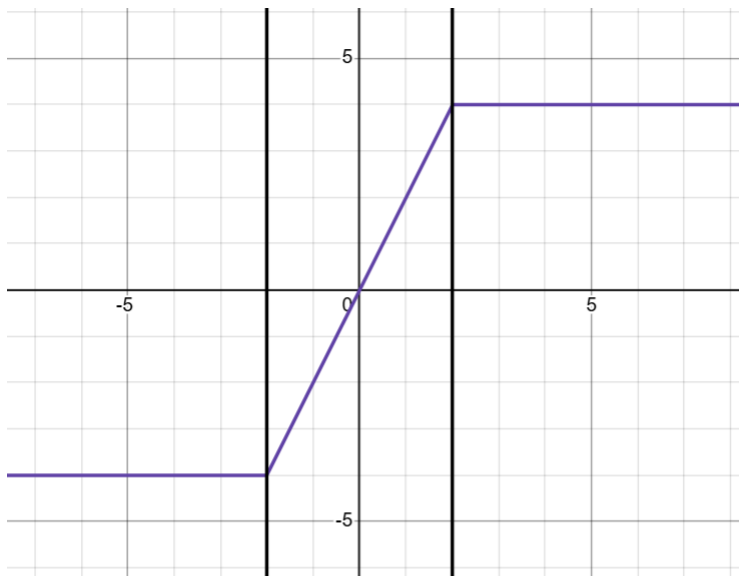
The maximum function will give back 2, and the minimum function will give back x, with the right-hand side giving $2x$

Solving this we would get $2-x = 2x$ which is $x=2/3$

Therefore, there are two real values of x for which the given equation holds.

Hence, 2 is the correct answer.

The graph of the function on the right hand side can be visualised as:



Hence, 2 is the correct answer.

VIDEO SOLUTION

55.9

Let us assume the amount invested to be P , and the rate of interest to be r .

Value of the investment after 3 years will be, $P(1+r)^3$

Value of the investment after 5 years will be, $P(1+r)^5$

$$(1+r)^2 = \frac{36}{25}$$

$$(1+r)^2 = 1.44$$

$$r = 0.2$$

We need to find the value of n for which $4000(1+r)^n > 20000$

$$(1+r)^n > 5$$

$$(1.2)^n > 5$$

We see that,

$$1.2^8 = 4.2999$$

$$1.2^9 = 5.15$$

Hence it takes 9 years to grow to over 20,000.

VIDEO SOLUTION

100+ CAT DILR Questions

cracku



56.A

Taking $10^x = a$

we get $a + \frac{4}{a} = \frac{81}{2}$

This would give the quadratic equation: $2a^2 - 81a + 8 = 0$

We want to find the sum of possible values of x , let the value of x be x_1 and x_2

these would correspond to $\log a_1$, and $\log a_2$

The sum of $\log a_1 + \log a_2$ would be $\log(a_1 \times a_2)$

From the quadratic equation we got above, we can see that the product of the possible values of a would be $8/2 = 4$

Therefore, the sum of values of x would be $\log(4)$ which would be $2 \log_{10} 2$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

57.A

Let us assume that the distance is D , speed of the train is S and time taken by the train is t .

t is nothing but $\frac{D}{S}$

Statement 1: Had the speed been 6 km per hour more, it would have needed 4 hours less

$$\frac{D}{S+6} = t - 4$$

$$\frac{D}{S+6} = \frac{D}{S} - 4$$

$$4 = \frac{D}{S} - \frac{D}{S+6}$$

$$\frac{S+6-S}{S(S+6)} = \frac{4}{D}$$

$$\frac{6}{S(S+6)} = \frac{4}{D}$$

$$D = \frac{2S(S+6)}{3}$$

Statement 2: Had the speed been 6 km per hour less, it would have needed 6 hours more

$$\frac{D}{S-6} = t + 6$$

$$D \left[\frac{1}{S-6} - \frac{1}{S} \right] = 6$$

$$\frac{S-S+6}{S(S-6)} = 6$$

$$D = S(S-6)$$

Equating the two equations for distance,

$$S(S-6) = \frac{2S(S+6)}{3}$$

$$3S - 18 = 2S + 12$$

$$S = 30$$

Hence the speed is 30 kmph

We know that the distance $D = S(S-6) = 30 \times 24 = 720$ km

VIDEO SOLUTION

58.2

Taking a log for each of the expressions, we get the following:

$$\log_3 4 = a, \log_4 5 = b, \log_5 6 = c, \log_6 7 = d, \log_7 8 = e, \log_8 9 = f$$

The expression $abcdef$ would then be: $\log_3 4 \times \log_4 5 \times \log_5 6 \times \log_6 7 \times \log_7 8 \times \log_8 9$

Next, we can use this property of log: $\frac{\log_b a}{\log_b c} = \log_c a$

Using this, we get:

$$\frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

All the terms will cancel out except: $\frac{\log 9}{\log 3} = \log_3 9 = 2$

Therefore, 2 is the correct answer.

VIDEO SOLUTION

59.A

Let us say the cost price of an item is X

It is said that it is marked to make a profit of 20%.

That means it is marked at 1.2X

Ravi gets a 10% discount on the marked price,

$$0.9(1.2X) = 1.08X$$

Saves 15 rupees, so $1.2X - 1.08X$

$$0.12X = 15$$

$$X = 125$$

Profit made by Gopi is $0.08(125) = 10$ rupees.

VIDEO SOLUTION

60.30

Let us assume the amount of Milk in the container to be X and the amount of water in the container to be Y.

We are told that $X+Y=300$.

Now, an operation is given where "an amount of this solution was taken out from the container which was twice the volume of water that was earlier poured into it, and water was poured to refill the container again"

Volume of the water initially is Y. If twice that amount is taken out, the percentage of the solution that is taken out will be, $\frac{2Y}{X+Y}$

That means the quantity of milk that will remain in the solution will be, $X \left(1 - \frac{2Y}{X+Y}\right)$

This value is given to be 72%, 72% of 300 will be 216

$$X \left(1 - \frac{2Y}{X+Y}\right) = 216$$

$$X \left(\frac{X+Y-2Y}{X+Y}\right) = 216$$

$$\text{Writing } X = 300 - Y$$

$$(300 - Y)(300 - 2Y) = 64800$$

Expanding this we have,

$$2Y^2 - 900Y + 25200 = 0$$

Factorising this equation we have,

$$2(Y - 30)(Y - 420) = 0$$

Y is either 30 or 420.

Given that the capacity of the container itself is 300, Y has to be 30.

Hence the amount of water initially is 30 Litres.

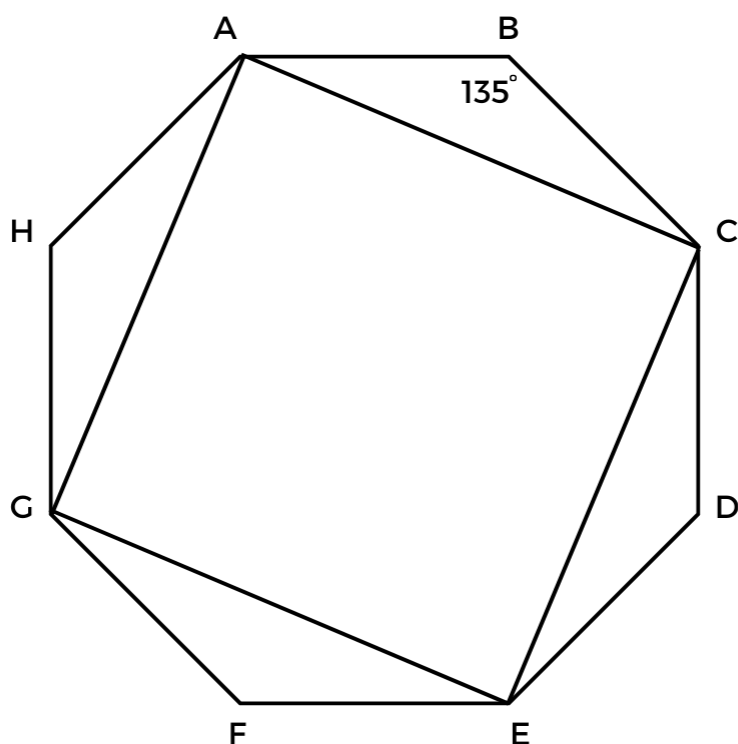
[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61.D

This is the figure in the question,



We are given that each side is 6cm long,

To find the side AC, we can use cosine rule, since we know each interior angle of the octagon(which is 135 degrees)

$$\cos(\angle ABC) = \frac{AB^2 + BC^2 - AC^2}{2(AB)(BC)}$$

$$\cos(135) = \frac{36 + 36 - AC^2}{2(36)}$$

$$-\frac{1}{\sqrt{2}} = \frac{72 - AC^2}{72}$$

$$AC^2 = 72 + 36\sqrt{2}$$

$$AC^2 = 36(2 + \sqrt{2})$$

Since AC is the side of the square, and the area of a square is square of the side.

Answer is $36(2 + \sqrt{2})$

[VIDEO SOLUTION](#)

62.8

The moduli will give out only non-negative outputs, and since we are to consider only integer values of x and y, this drastically reduces the possible cases.

We can get 2 from either $2+0$ or $1+1$

We get a $2+0$ form when either the first term or the second term is 0

The second term is 0; this is when $x=y$, in this case, $|2x|=2$, where x can be 1 or -1; therefore, the two cases are (1,1) and (-1,-1)

The first term is 0; this is the case when $x = -y$, in this case, $|x - (-x)|=2$, giving $x=1$ or -1 yet again, here the two cases are (1,-1) and (-1,1)

The other way we can get 2 is through $1+1$

This is possible when one of the terms is 0; if $y=0$, $|x|+|x|=2$, where x can be 1 or -1, giving two cases (1,0) and (-1,0)

Similarly, for $x=0$, we get two cases, (0,1) and (0,-1)

Therefore, there are 8 pairs of (x,y) that satisfy the given equation.

[VIDEO SOLUTION](#)

63. C

We are given, $f(x) + 2f\left(\frac{1}{x}\right) = 3x$

Substituting $\frac{1}{x}$ for x

$$f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}$$

Multiplying the second equation by 2 we will have

$$2f\left(\frac{1}{x}\right) + 4f(x) = \frac{6}{x}$$

Subtracting the first equation from the second equation we have,

$$3f(x) = \frac{6}{x} - 3x$$

$$f(x) = \frac{2}{x} - x$$

We want the sum of values when this function equals 3,

$$\frac{2}{x} - x = 3$$

$$x^2 + 3x - 2 = 0$$

Since the discriminant is greater than zero, both values of x will be real, and we can directly take the sum of values of x to be,

$$-\frac{3}{1}$$

Answer is -3.

[VIDEO SOLUTION](#)

64. B

Arranging the equation, we know that there are no solutions when the lines are parallel:

for that the condition had to be $\frac{p}{3} = -\frac{4}{k} \neq \frac{2}{a}$

Checking through options:

Option A: using the first and last terms in our relation, we see that ap must not be equal to 6 for the lines to be parallel. This option puts no conditions on that and thus is not relevant.

Option C: This question is the opposite of what we want; if this is true, the lines can never be parallel.

Option D: Using the first and second terms of the relation, we see that we want $kp = -12$, or $kp-12 = 0$. Hence, this statement is not what we want.

Option B: Using the second and third terms, we see that we do not want k equals to $-2a$, or we do not want $k+2a$ to be equal to 0

Therefore, B is a condition that is necessary for the lines to be parallel and have no solution.

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

65. B

We are told that Rajesh manages 20 hectares and Vimal manages 30 hectares

For Vimal, we know the distribution of the land between Wheat and Mustard, 5:3

So, wheat area will be, $\frac{5}{8}(30)$

Mustard area will be, $\frac{3}{8}(30)$

Similarly, let us assume that the distribution of crops between Wheat and Mustard to be $k:1$

Wheat will be, $\frac{k}{k+1}(20)$

Mustard will be, $\frac{1}{k+1}(20)$

We are told that total area of Wheat and Mustard is in the ratio 11:9

Adding them up we get,

$$\frac{\left(\frac{150}{8} + \frac{20k}{k+1}\right)}{\left(\frac{90}{8} + \frac{20}{k+1}\right)} = \frac{11}{9}$$

$$\frac{\left(\frac{15}{8} + \frac{2k}{k+1}\right)}{\left(\frac{9}{8} + \frac{2}{k+1}\right)} = \frac{11}{9}$$

$$\frac{135}{8} + \frac{18k}{k+1} = \frac{99}{8} + \frac{22}{k+1}$$

$$\frac{36}{8} = \frac{22-18k}{k+1}$$

$$44 - 36k = 9k + 9$$

$$45k = 35$$

$$k = \frac{7}{9}$$

Hence the ratio of distribution of area between Wheat and Mustard for Rajesh is $\frac{7}{9}$

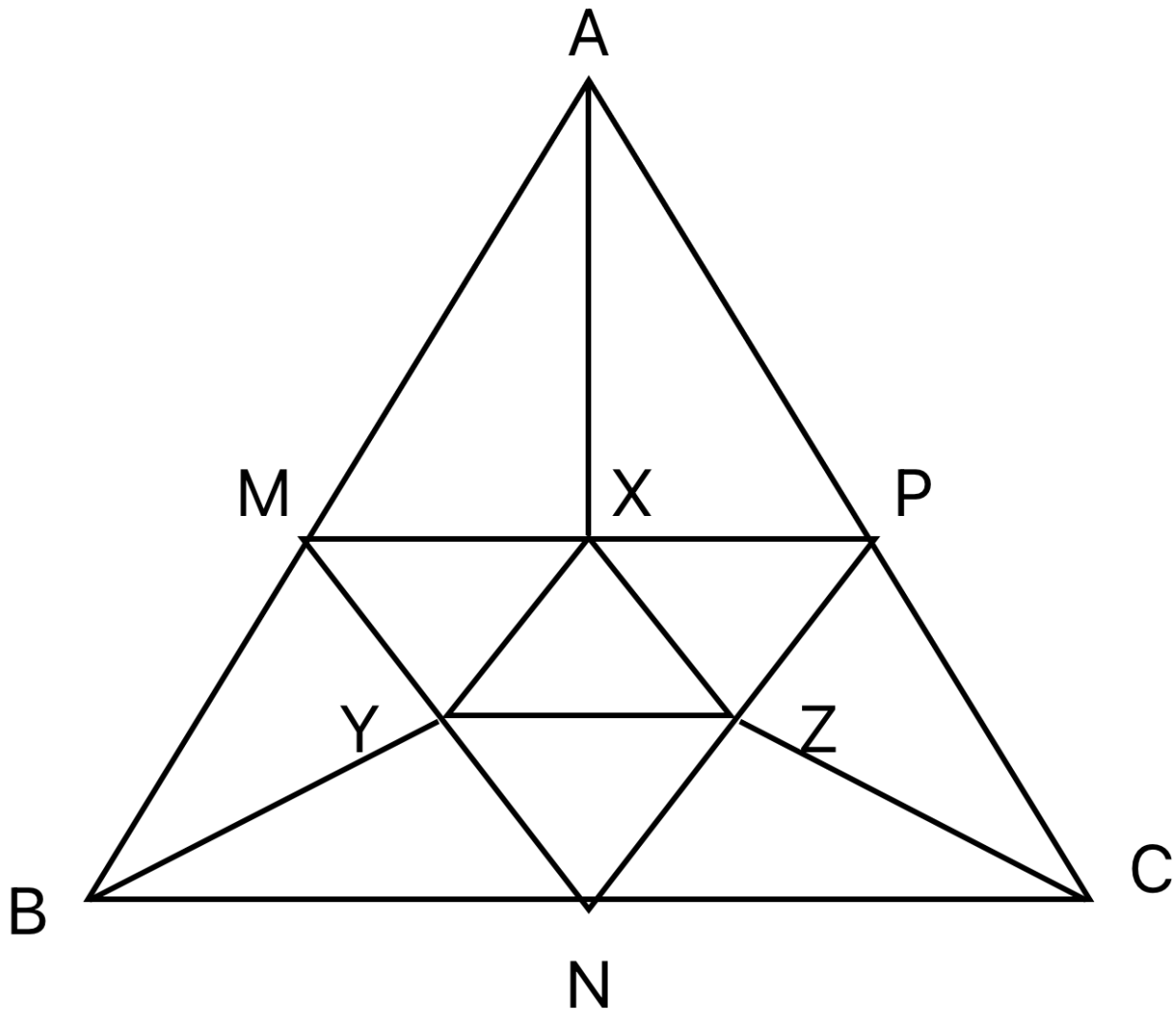
[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66.90

The question describes a figure of the following nature,



We are told that the area of the triangle ABC is 1440.

By proportionality theorem, the area of MPN should be one fourth of that.

Since we know that, $\frac{\text{Area } \triangle_1}{\text{Area } \triangle_2} = \frac{(\text{side of } \triangle_1)^2}{(\text{side of } \triangle_2)^2}$

And since it is given that MPN are midpoints of the respective sides, area of triangle MPN is $\triangle \frac{ABC}{4} = \frac{1440}{4} = 360$

Now, the medians from each side intersect, MP MN and NP are X Y and Z respectively, since these sides are proportional to the main triangle, the point of intersection of the medians with these sides will divide the side MP, MN and NP in half.

Then, Area of Triangle XYZ will be one fourth the area MPN,

$$\triangle \frac{MPN}{4} = \frac{360}{4} = 90$$

The answer is 90.

[VIDEO SOLUTION](#)

67.378

We need to consider single-digit, two-digit and three-digit numbers

Single digit: There are only nine numbers for a single-digit number (not including zero since we are looking for positive integers only)

Double digits: The ten's digit can be chosen in 9 ways (not including 0), and the unit digit can be chosen in 9 ways (including 0 but excluding the digit used in the ten's place). Hence, a total of 81 numbers.

Three-digit: The hundred's digit can be 1, 2, 3 or 4; hence, there are four options for the hundred's digit; for the ten's digit, there will be 9 options (numbers from 0 to 9 except the one chosen in the hundred's place); for unit's digit there would be 8 options. Hence, a total of 288 numbers

Giving the total numbers as $288 + 81 + 9 = 378$

Therefore, 378 is the correct answer.

[VIDEO SOLUTION](#)

68. C

We are told that there was two successive increments in the salary, with the second increment percentage twice the first one. Total Increment was 187.5%.

Drawing up the equation

$$(1 + z)(1 + 2z) = 1.875$$

$$1 + 3z + 2z^2 = 1.875$$

$$2z^2 + 3z - 0.875$$

$$z = \frac{-3 \pm \sqrt{9 + 8(0.875)}}{4}$$

$$z = \frac{-3 \pm 4}{4}$$

$$z = 0.25$$

Answer is 25%.

[VIDEO SOLUTION](#)